# Industrial Policy in a Globalized Economy
## Domestic Embeddedness and Industrial Subsidy in the United States

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September 8, 2026

*Writing Sample Prepared for 2026-27 Job Market Applications*

**Abstract**

Governments routinely subsidize foreign firms, even as industrial policy is justified in terms of national economic development. Why do governments support firms whose benefits may accrue abroad, and what determines how much support they receive? I argue that subsidy allocation depends not simply on firm nationality but on firms' domestic embeddedness, the extent to which their activities generate visible economic benefits within the domestic economy. Using firm-level data on U.S. subsidies from 2000–2023, I show that foreign firms receive fewer and substantially smaller subsidies than U.S. firms, but this disadvantage is substantially attenuated among labor-intensive firms. By contrast, reliance on foreign suppliers does not reduce government support, and supplier labor intensity does not condition this relationship. These findings show that governments distinguish among dimensions of firm foreignness: firm nationality carries a clear penalty, while foreign supplier reliance does not. Industrial policy thus provides a means of navigating economic interdependence by directing support toward firms that generate politically attributable benefits at home.

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# Introduction

Industrial policy has re-emerged as a core instrument of state economic strategy amid renewed concerns over competitiveness, technological security, and geopolitical rivalry (Juhász, Lane, and Rodrik, 2024; Allan and Nahm, 2024; Breznitz and Gingrich, 2025). Across advanced economies, governments are channeling unprecedented subsidies and tax incentives to promote domestic production and secure critical industries (Lane, 2020; Naughton, 2021).

Yet in a deeply globalized economy, the firms receiving industrial support are rarely confined within national borders (Osgood, 2018; Kim and Osgood, 2019). Many subsidized firms are foreign or rely on global supply chains, even when production occurs domestically (Dür, Eckhardt, and Poletti, 2019; Gereffi, 2013). This reality creates a fundamental puzzle: despite industrial policy being justified in terms of national economic development, a substantial share of government subsidies flow to foreign firms operating across borders (Ding, 2026). Why do governments support firms whose benefits may accrue abroad, and what determines how much support they receive?

I argue that governments allocate industrial subsidies based on firms' degree of domestic embeddedness. Because industrial policy is typically justified in terms of local economic development, policymakers have incentives to support firms whose activities generate visible and locally attributable benefits (Rickard, 2018; Jensen and Malesky, 2018; Bartik, 2019). Foreign nationality and global integration complicate this logic by weakening the link between government support and domestic economic gains. When firms' operations are dispersed across borders, the benefits of subsidies become harder to attribute to domestic constituencies, reducing their political value (Lee, 2022, 2023; Tingley et al., 2015; Chilton, Milner, and Tingley, 2017). Firms with stronger domestic ties, by contrast, produce more visible and politically attributable contributions, making them more attractive targets for government support (Tingley et al., 2015; Chilton, Milner, and Tingley, 2017; Jensen, Findley, and Nielson, 2020; Slattery, 2023).

Yet foreign nationality does not uniformly disadvantage firms. I argue that domestic economic contributions, particularly employment generation at the firm level, can offset concerns associated with foreign nationality. Labor-intensive firms generate visible and locally concentrated benefits, making subsidies more politically defensible even for foreign firms. By contrast, economic activity embedded in upstream supply chains, especially abroad, is less observable and harder to attribute to policymakers. As a result, while firm-level employment can mitigate the disadvantages associated with foreign nationality, foreign supply-chain linkages are less likely to shape subsidy allocation.

To evaluate these expectations, I assemble a firm-level dataset of U.S. industrial subsidies from 2000 to 2023, combining information on subsidy awards with firm nationality, supply chains, and labor intensity. The United States provides a useful setting: it is highly integrated into global production networks while also maintaining extensive industrial support programs across federal, state, and local governments (Kose et al., 2017; Slattery, 2025).

The analysis reveals three broad findings that together explain why and when governments support foreign firms. First, while foreign firms are systematically less likely to receive subsidies and receive smaller amounts when they do, a substantial share of industrial support nonetheless flows to foreign firms, a pattern explained by the remaining findings. Second, the disadvantage associated with foreign nationality diminishes significantly among labor-intensive firms. This pattern is consistent with the argument that an employment-generating orientation can make support for foreign firms more politically defensible, although the available measure does not identify where firms' employees are located. Third, reliance on foreign suppliers does not reduce subsidy receipt, whether among U.S. or foreign firms, and supplier labor intensity does not condition this relationship, indicating that global supply-chain integration is evaluated differently from foreign nationality. Together, these findings show that governments do not simply penalize foreignness, but selectively reward foreign firms whose characteristics signal potentially visible economic benefits.

This study contributes to several strands of research. First, it advances work on the distributive politics of industrial policy by showing that subsidy allocation varies systematically across firms within industries, not only across sectors or regions (McGillivray, 2004; Rickard, 2018; Slattery, 2023). Second, it speaks to debates on industrial policy in a globalized economy by demonstrating how governments manage, rather than retreat from, economic interdependence, prioritizing firms that translate global integration into visible domestic benefits (Impullitti, 2010; Kondo, 2013; Farrell and Newman, 2019). Third, it extends firm-centered approaches in international political economy by identifying domestic embeddedness, particularly firm-level employment, as the key mechanism explaining when and why states direct support toward globally integrated firms (Melitz, 2003; Kim and Osgood, 2019; Lee and Liou, 2022).

More broadly, the findings highlight a shift in how governments navigate globalization through industrial policy. Rather than simply favoring nationally based firms, policymakers prioritize forms of economic activity that are both globally integrated and domestically legible. This reflects a fundamental constraint of contemporary industrial policy: governments depend on globally embedded firms to achieve strategic objectives, yet must justify support in terms of visible domestic benefits. Industrial policy thus operates not by excluding foreign firms, but by selectively rewarding those whose activities generate politically attributable gains at home.

# Theory

## Industrial policy: definitions, motivations and distributive logic

***Definition of industrial policy*** Industrial policy refers to state-led efforts to shape economic activity in pursuit of public objectives, often through targeted support for particular firms, sectors, or technologies (Lane, 2020; Juhász, Lane, and Rodrik, 2024).[1] Industrial policy has a long history, from postwar reconstruction strategies in Europe and Japan to the developmental states of East Asia. In recent years, it has re-emerged at the center of economic statecraft, illustrated by initiatives such as the U.S. Inflation Reduction Act (IRA) and CHIPS and Science Act (Muresianu, 2024), as well as China's Made in China 2025 strategy (Kawase, 2022).

Governments implement industrial policy through a wide array of instruments, including production subsidies such as direct grants, tax incentives, concessional loans, and infrastructure support; consumption subsidies that encourage demand for domestically produced goods; tariffs and trade barriers; and broader forms of state coordination, including strategic planning and public R&D investment. These tools vary in how directly they intervene in markets and in whether they target producers, consumers, or entire industries. This study focuses on production subsidies, which represent one of the most direct and politically visible forms of state support.

Although industrial policy is often framed as sectoral policy, many interventions are implemented at the firm level through location-specific tax incentives, customized financial packages, or public infrastructure commitments. Governments frequently negotiate directly with individual firms, particularly large or strategically important ones, making firm-level allocation central to understanding the distributive politics of industrial policy. Examining industrial policy at the firm level therefore provides a valuable lens for analyzing how states prioritize and support specific economic actors (Jansa and Gray, 2016; Slattery, 2023).

[1] The OECD defines industrial policy as government support aimed at enhancing or restructuring specific economic activities, frequently by targeting firms based on activity, technology, location, size, or age. Such policies seek to address challenges beyond market capacities, including climate change mitigation and supply-chain resilience.

***Motivations and distributive logic*** Industrial policy serves both economic and political objectives. Economically, governments deploy subsidies, tax incentives, and infrastructure investments to stimulate growth, create jobs, and enhance the global competitiveness of firms. These efforts have intensified under globalization, as states compete for mobile capital while seeking to secure critical industries amid shifting trade dynamics and geopolitical tensions (McGillivray, 2004; Rickard, 2018; Impullitti, 2010; Baldwin and Krugman, 2004; De Bièvre and Poletti, 2020; Carnegie and Gaikwad, 2022). Subsidies can reduce firms' costs, enable investment and expansion, and enhance productivity, particularly in strategic sectors (Aghion et al., 2015; Kalouptsidi, 2017; Rotemberg, 2019).

Industrial policy also operates as a powerful instrument of distributive politics. Subsidies allow policymakers to deliver targeted benefits to firms, workers, and communities in politically salient regions or sectors, generating electoral rewards through visible economic outcomes such as job creation or new investment (Dewatripont and Seabright, 2006; Rickard, 2018; Slattery, 2023; De Bièvre and Dür, 2005; Menon and Osgood, 2024). Firms and other beneficiaries may reciprocate through political support, lobbying, or public endorsement (Drope and Hansen, 2006; Jansa and Gray, 2016; Zhang, 2025). As a result, governments evaluate subsidy allocation not only in terms of aggregate economic efficiency but also in terms of domestic political returns.

In a globalized economy, however, this distributive logic becomes more complex. Firms increasingly operate across borders, such that the economic benefits of subsidies may also extend beyond national boundaries. This creates genuine uncertainty for policymakers about whether the resulting gains will accrue domestically, complicating the justification for industrial support. Why do governments support firms whose benefits may accrue partly abroad, and what determines how much support these firms receive?

The answer, I argue, lies in how governments evaluate the domestic embeddedness of firms, the extent to which their activities generate visible and locally attributable economic benefits. Foreignness is not a single condition but varies across multiple dimensions, most notably firm nationality and supply-chain linkages, which differ in how directly they connect firms' activities to the domestic economy and in how visible their benefits are to policymakers and constituents. Understanding how governments weigh these dimensions is central to explaining the distributive politics of industrial policy in a globalized economy.

## Firm foreignness and subsidy allocation

The globalization of production blurs the distinction between U.S. and foreign firms, complicating policymakers' efforts to identify which firms advance national economic interests. Yet governments continue to direct substantial support toward foreign firms and firms embedded in global supply chains, raising the question of what determines whether and how much support these firms receive. I conceptualize firm foreignness along two primary dimensions: firm nationality and supply chains. These dimensions differ not only in how firms are economically embedded across borders, but also in how visible and politically attributable their activities are to policymakers and domestic constituencies. Even firms operating domestically may differ substantially in the extent to which their economic activities are embedded abroad, shaping both the economic and political calculus of subsidy allocation.

***Firm nationality.*** One of the most visible dimensions of firm foreignness is nationality (Zaheer, 1995). As a baseline condition, foreign firms face a systematic disadvantage in the allocation of industrial subsidies. Policymakers may be less willing to support foreign firms because the benefits of subsidies are perceived as more likely to flow abroad and are less clearly attributable to domestic constituencies, generating weaker political returns than support for U.S. firms (Feng, Kerner, and Sumner, 2019).

A key mechanism behind this penalty is political reciprocity. U.S. firms are better positioned to reward policymakers through campaign contributions, lobbying, and local economic engagement, while also embedding themselves in domestic constituencies through jobs and community investment. Foreign firms, even when operating through U.S. subsidiaries, face legal and practical constraints in electoral politics (Siegel, 2007; Johns and Wellhausen, 2021; Lee, 2022; Kim and Siegel, 2023). These constraints reduce their political value, lowering their likelihood of receiving subsidies even when their economic contributions are comparable to those of U.S. firms.

The skepticism is not merely theoretical. When Honda became the first Japanese automaker to open a U.S. assembly plant in Marysville, Ohio, in 1982, contemporary critics warned that such investment risked turning American industry into what one account called manufacturing colonies of foreign firms. The concern centered on Honda's Japanese national affiliation rather than on what the plant produced (Behr, 1987; Williams, 2022).

**Hypothesis 1.** Foreign firms are less likely to receive industrial support than U.S. firms.

***Supply-chain nationality.*** A second dimension of foreignness lies in a firm's supply chain, specifically whether it sources inputs domestically or internationally. In principle, firms that rely more heavily on foreign suppliers may be less likely to receive industrial support than those embedded in domestic supply chains. When subsidies expand production, upstream suppliers often share in those gains, implying that reliance on foreign suppliers can diffuse the economic effects of industrial policy beyond national borders (Johns and Wellhausen, 2016; Zeng, Sebold, and Lu, 2018).

However, unlike firm nationality, supply-chain linkages are less directly observable and more difficult to attribute to policy decisions. The economic benefits associated with suppliers are often indirect and dispersed across multiple firms and locations, making them less salient for voters and policymakers. As a result, while governments may express concerns about global supply-chain dependence, these considerations are likely to play a more limited role in subsidy allocation than more visible firm-level characteristics. Even prominent recent cases illustrate this asymmetry: congressional opposition to Chinese automaker BYD's prospective U.S. plants has centered primarily on the firm's Chinese national affiliation and related national-security concerns, while foreign-sourced batteries and parts have featured as a secondary argument (Office of Senator Tammy Baldwin, 2024; Reuters, 2026).

**Hypothesis 2.** Firms that rely more heavily on foreign suppliers receive less industrial support than firms embedded in domestic supply chains.

## Domestic economic contributions: labor intensity as a moderator

While foreign nationality systematically disadvantages firms in subsidy allocation, not all foreign firms are treated equally. The central question is what allows some foreign firms to overcome this baseline penalty and receive government support. I argue that the answer lies in the domestic economic contributions firms generate, particularly employment. A central characteristic shaping these contributions is labor intensity, the extent to which production relies on labor relative to capital (Helpman, Melitz, and Yeaple, 2004; Pandya, 2010).

In democratic systems, policymakers are highly responsive to employment outcomes, often citing job creation as a central justification for industrial policy (Lee and Osgood, 2018; Slattery, 2023; Jensen, Findley, and Nielson, 2020). Subsidies are easier to defend politically when they produce visible local jobs, benefits that voters understand and value more readily than abstract productivity gains (Owen, 2017; Owen and Johnston, 2017). Labor-intensive firms therefore provide policymakers with a visible, defensible justification for government support, one that is harder to construct around capital-intensive activity.

This visibility is especially consequential for foreign firms. Foreign nationality raises concerns about economic leakage and national alignment, and policymakers need some countervailing evidence of domestic benefit to justify support despite these concerns. Visible job creation supplies exactly that evidence: foreign firms that employ large numbers of local workers can point to a concrete, attributable domestic contribution, allowing them to be perceived as contributing meaningfully to the domestic economy even when headquartered abroad.

By contrast, capital-intensive foreign firms offer fewer visible domestic benefits. Their contributions to the domestic economy, whether through productivity gains, technology transfer, or capital formation, are diffuse and difficult for policymakers to point to as evidence of local benefit. Where labor-intensive firms generate a concrete, countable metric (jobs), capital-intensive firms generate value that is harder to attribute to the domestic economy specifically, since returns to capital can plausibly be construed as accruing abroad rather than to domestic workers or communities. This combination of low visibility and weak attributability leaves capital-intensive foreign firms with little to offer policymakers seeking to justify support, and they are more likely to face exclusion from it.

For instance, BMW's Spartanburg, South Carolina, plant has faced little sustained political controversy despite BMW's status as a German firm and is now credited with supporting some 11,000 jobs statewide through its supplier network, a multiplier effect state officials and BMW itself have repeatedly invoked to justify the original incentive package (BMW Group, 2024; Printz, 2026). Wisconsin's 2017 deal with Foxconn shows what happens when the same logic fails: the state structured its up to $2.85 billion package explicitly around annual hiring thresholds, but when the promised 13,000 jobs shrank to roughly 1,500, the deal became a national symbol of subsidy failure and most of the committed funds were clawed back (Reuters, 2021; Burke, Feng, and Neely, 2022; Schulz, 2024). Capital-intensive investment struggles to generate the same justification even at far larger scale: TSMC's Arizona investment illustrates the ambiguity surrounding capital-intensive projects (Hayashi and Jie, 2023; O'Marah, 2023; Byung-soo, 2025). Although the project is expected to create thousands of manufacturing and construction jobs, its employment effects remain modest relative to the scale of the investment and public support, making job creation a less straightforward political justification than in more labor-intensive projects.

**Hypothesis 3.** The disadvantage faced by foreign firms is attenuated among more labor-intensive firms.

***Supplier labor intensity.*** Labor intensity may also operate through supply chains. Firms connected to labor-intensive foreign suppliers could, in principle, shift employment benefits abroad, raising concerns about whether subsidies support domestic job creation (Malesky and Mosley, 2018). However, as with supply-chain nationality more broadly, these dynamics are less directly observable and more difficult to attribute to policy decisions than firm-level employment.

Because policymakers and voters respond most strongly to visible, firm-level employment, supplier labor intensity is likely to carry less political weight. Nevertheless, when supply-chain employment is considered, reliance on labor-intensive foreign suppliers should heighten concerns that subsidies support jobs abroad rather than at home. The CHIPS Act debate over TSMC illustrates the limited visibility of these effects: although advocates emphasized the project's indirect supplier and construction jobs, these benefits featured less prominently in political debate than employment created directly by the subsidized firm.

**Hypothesis 4.** The negative relationship between reliance on foreign suppliers and subsidy allocation is stronger when those suppliers are more labor-intensive.

Taken as a whole, this framework explains when and why governments support foreign firms despite the political costs of doing so. The baseline disadvantage associated with foreign nationality is conditional on the domestic economic benefits firms generate. Foreign firms that create substantial domestic employment can overcome political barriers by producing visible and locally attributable contributions, while the implications of global supply-chain linkages are more contingent and less central to subsidy decisions. Industrial policy thus operates not by excluding foreign firms, but by selectively rewarding those whose activities generate politically attributable gains at home.

# Data and Research Design

To evaluate the theoretical expectations, I analyze industrial subsidies in the United States, which offer several analytical advantages. As one of the most globalized economies, with high levels of cross-border investment, trade, and multinational firm activity, the United States offers substantial variation in firms' foreignness and domestic economic contributions (Kose et al., 2017). At the same time, governments at the federal, state, and local levels actively deploy subsidies to attract investment and shape industrial development, creating meaningful variation in policy allocation across firms (Jensen, Findley, and Nielson, 2020). These features make the United States well suited for examining how governments distribute industrial policy benefits in a globalized economy.

Because the theory concerns how governments allocate subsidies to individual firms, the unit of analysis is the firm-year. To construct the dataset, I integrate three primary sources: Compustat, Orbis, and Subsidy Tracker. Compustat and Orbis provide detailed financial and organizational information on publicly traded firms, while Subsidy Tracker offers comprehensive records of government subsidies awarded to firms.[2] The final dataset covers publicly traded firms from 2000 to 2023, a period during which globalization and industrial policy both intensified.

## Industrial support for firms

The dependent variables measure the industrial subsidies firms receive from government entities.[3] I use two indicators: the number of subsidies awarded to a firm in a given year and the total monetary value of subsidies received.[4] Because both outcomes include zero values and are highly right-skewed, I transform them as $\log(1 + \text{count})$ and $\log(1 + \text{amount})$ in the regression analyses.

Figure 1 illustrates the geographic distribution of firms in the sample and their subsidy receipt. Firms are located throughout the United States, reflecting the nationwide scope of industrial policy across federal, state, and local governments. Both subsidized and non-subsidized firms appear across regions, industries, and local economies, indicating that subsidy allocation does not appear to account for the observed variation. This broad geographic coverage provides substantial variation in subsidy receipt and motivates the central question of this study: which firm characteristics make some firms more likely than others to receive industrial support? The following section therefore turns to the key explanatory variables capturing firms' foreignness.

## Firm foreignness

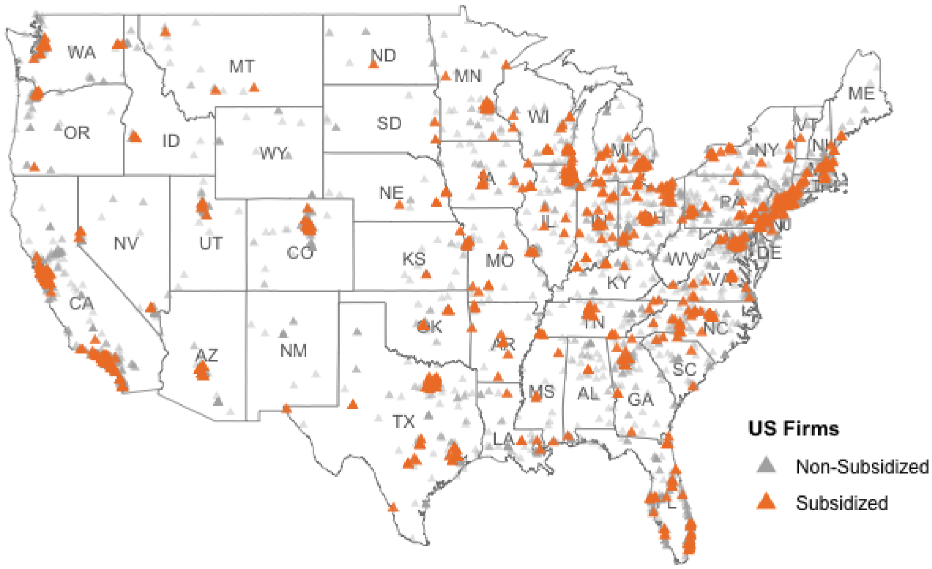
The core independent variables capture firm foreignness along two dimensions: firm nationality and supply chains. Firm nationality and supply-chain nationality capture distinct

[2] Because these datasets use different firm identifiers, I matched records using GVKEY, ISIN, TIC, and company names through a combination of automated matching and manual verification using the WRDS Capital IQ ID web query.

[3] Industrial subsidy activity varies over time, with both subsidy counts and total amounts peaking in 2013. See Appendix Figure A1 for annual trends.

[4] Subsidies include grants, tax incentives, subsidized loans, and infrastructure support provided by federal, state, and local governments. The allocation varies across levels of government: state governments account for 73.3% of subsidy projects, while federal programs account for 54.3% of total subsidy dollars. Although subsidy decisions are made by different governmental actors, the theoretical argument developed here concerns the political implications of firm foreignness, which are not unique to any single level of government.

Figure 1: Geographic Distribution of Firms and Industrial Subsidy Receipt




Note: Points represent firm-year observations located at firms' headquarters. Orange points indicate firms receiving at least one industrial subsidy during the observation year; gray points indicate firms receiving no subsidy.

dimensions of firms' connections to the domestic economy. Whereas firm nationality identifies the country in which the firm is headquartered, supply-chain nationality measures where production inputs are sourced. These dimensions frequently diverge. For example, some U.S. firms rely heavily on foreign suppliers, while some foreign firms source much of their procurement domestically. Distinguishing between firm nationality and supply chains allows me to evaluate whether governments respond primarily to firms' legal nationality or to where their economic activity takes place.

***Firm nationality.*** I measure firm nationality using headquarters country and fiscal country information from Compustat, cross-validated against firm-nationality information from Orbis. The two measures coincide in the matched sample. I construct a binary indicator, Foreign firm, coded one if a firm is headquartered outside the United States and zero otherwise. Throughout the analysis, I refer to these firms as foreign firms and to firms headquartered in the United States as U.S. firms. Firm nationality represents the most visible dimension of foreignness and corresponds directly to Hypothesis 1.

Table 1 shows that foreign firms are consistently less likely to receive industrial subsidies than comparable U.S. firms. While foreign firms account for 23.0% of all firm-years, they

Table 1: Subsidy Receipt by Firm Nationality

| | All firm-years | U.S. firms | Foreign firms |
|---|---|---|---|
| N | 151,338 | 116,456 | 34,882 |
| Share of sample | 100% | 77.0% | 23.0% |
| Share of all recipients | — | 84.6% | 15.4% |
| % receiving any subsidy | 9.2% | 10.1% | 6.2% |
| Avg. # awards among recipients | 4.14 | 4.22 | 3.69 |
| Mean subsidy amount (recipients only) | — | $12,861,464 | $11,119,965 |

Note: Recipient-conditional statistics (rows 5–6) computed on firm-years with *sub_count* > 0.

represent only 15.4% of subsidy recipients.[5] Put differently, 10.1% of U.S. firm-years receive at least one subsidy, compared to only 6.2% of foreign firm-years. Among firms receiving support, U.S. firms also receive slightly more awards on average (4.22 versus 3.69) and somewhat larger subsidy amounts. These descriptive differences are consistent with the expectation that foreign nationality is associated with lower access to industrial subsidies. Firm nationality, however, captures only one dimension of firm foreignness. To examine whether governments also respond to firms' production networks, I next measure the nationality of firms' supply chains.

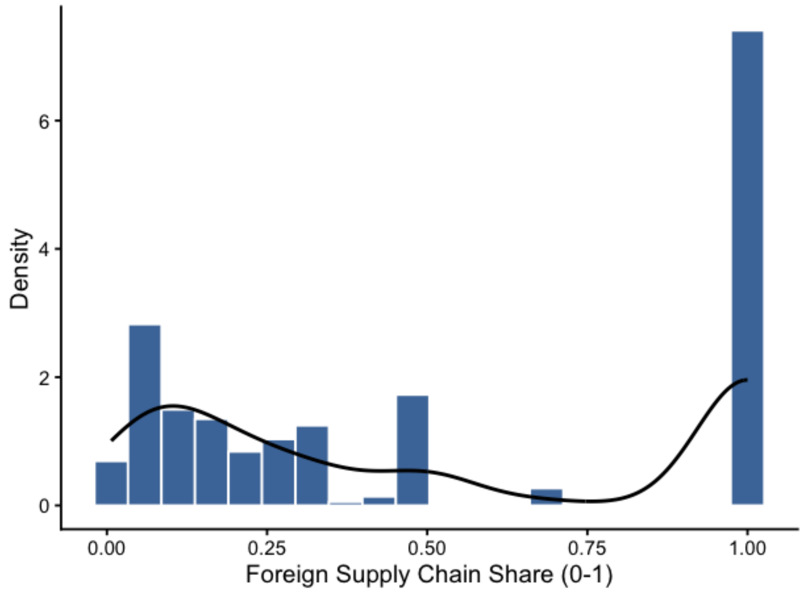
***Supply chain nationality.*** To capture firms' integration into global production networks, I construct *Foreign supplier share* using Compustat Segment data on major suppliers and their geographic location. This variable measures the proportion of a firm's procurement sourced from foreign upstream suppliers, weighted by transaction volume.[6] Approximately 92% of firm-year observations report no foreign suppliers and therefore have a foreign supplier share of zero. Among the remaining 8%, however, reliance on foreign procurement varies substantially, ranging from limited foreign sourcing to firms whose reported major suppliers are almost entirely foreign (Figure 2). This heterogeneity captures the extent to which subsidies may generate economic spillovers abroad rather than within the domestic economy and corresponds to Hypothesis 2.

Although related, firm nationality and supply chain nationality capture distinct dimensions of foreignness. A U.S. firm may rely heavily on foreign suppliers, while a foreign firm

[5] Among foreign firms, recipients are headquartered in 45 countries. Japanese firms receive the largest number of subsidies, followed by firms from Germany, Switzerland, and Canada.

[6] Supplier data are derived from SEC disclosures and linked through verified name-matching procedures (Patatoukas, 2012; Cohen and Li, 2020).

Figure 2: Distribution of foreign supply chain share (non-zero observations)




Note: The figure displays the distribution of Foreign supplier share among firms with positive foreign supplier shares only (8% of firm-year observations). Approximately 92% of firm-year observations report no foreign suppliers and are omitted for visual clarity.

may source much of its procurement domestically. Distinguishing between these dimensions allows me to assess whether governments respond primarily to firms' headquarters country or to the geographic location of firms' economic activity.

## Labor intensity

To approximate firms' employment-generating orientation, I measure labor intensity at both the firm and supply-chain levels. These measures capture the importance of labor relative to firms' economic scale, but they do not identify where employees are located. Accordingly, I interpret labor intensity as a proxy for employment orientation rather than as a direct measure of domestic employment.

***Firm labor intensity.*** Firm-level labor intensity is measured as the ratio of employees to revenue using Compustat data.[7] Higher values indicate that a firm relies more heavily on labor relative to its economic scale. This measure corresponds to Hypothesis 3. Because employment is generally more visible and politically salient than capital investment, greater labor intensity may mitigate the disadvantage associated with foreign nationality. However,

[7] Revenue is measured in millions of U.S. dollars.

because the measure does not identify employees' geographic location, the results should be interpreted as evidence about firms' employment-generating orientation rather than as direct evidence of domestic employment.

***Supplier labor intensity.*** I measure the labor intensity of firms' upstream supply chains using firm-level data on supplier companies. Specifically, I calculate each supplier firm's labor intensity as the ratio of employees to revenue and merge this information with focal firms based on observed supplier relationships. Higher values indicate that a focal firm is connected to more labor-intensive production through its upstream supply chain. This measure corresponds to Hypothesis 4. If policymakers view reliance on labor-intensive foreign suppliers as indicating that employment benefits accrue abroad, the negative relationship between foreign supplier reliance and subsidy allocation should become stronger as supplier labor intensity increases. Because supplier employment is less visible and attributable than employment at the focal firm, however, this moderating relationship may be comparatively weak.

## Control variables

The analysis includes several control variables capturing baseline firm characteristics that may influence subsidy allocation. FFirm size is a composite measure constructed by standardizing annual revenue and employee count and averaging the two standardized components. Additional controls include industry fixed effects and year fixed effects to account for sectoral differences and temporal variation in industrial policy. These controls help isolate the relationship between firm foreignness, labor intensity, and subsidy allocation.

In sum, these data allow for a comprehensive test of how firm nationality, supply chains, and employment characteristics jointly shape governments' allocation of industrial subsidies across firms.

## Model specification

To evaluate the theoretical expectations, I estimate regression models at the firm-year level. The two dependent variables are the number of subsidies awarded to a firm and the total monetary value of those subsidies. Because both outcomes contain zeros and are highly right-skewed, I transform them as follows:

$$Y_{i,t}^{(k)} = \log\left(1 + \text{Subsidy}_{i,t}^{(k)}\right), \qquad k \in \{\text{count}, \text{amount}\},$$

where $i$ indexes firms, $t$ indexes years, and $k$ denotes the subsidy outcome.

The baseline models examine how firm nationality and foreign supplier reliance are associated with subsidy allocation. The two dimensions of foreignness are first entered separately and then jointly, as represented by the following general specification:

$$\begin{aligned} Y_{i,t}^{(k)} = \beta_0 &+ \beta_1 \text{Foreign firm}_{i,t} + \beta_2 \text{Foreign supplier share}_{i,t-1} \\ &+ \boldsymbol{\gamma}' \mathbf{Z}_{i,t} + \alpha_t^{\text{Year}} + \alpha_{s(i,t)}^{\text{State}} + \alpha_{j(i,t)}^{\text{Industry}} + \varepsilon_{i,t}, \end{aligned}$$

where $\mathbf{Z}_{i,t}$ contains the firm-size control, constructed from standardized annual revenue and employee count. The models include year fixed effects to account for temporal changes in industrial subsidy activity, state fixed effects to account for geographic differences in subsidy allocation, and four-digit NAICS industry fixed effects to account for sectoral heterogeneity. Here, $s(i,t)$ and $j(i,t)$ denote the state and industry associated with firm $i$ in year $t$. Hypothesis 1 predicts a negative coefficient on foreign firm, while Hypothesis 2 predicts a negative, although potentially weaker, coefficient on foreign supplier share.

I next examine whether firms' employment orientation conditions these relationships. The firm-level interaction model evaluates whether labor intensity moderates the disadvantage associated with foreign nationality:

$$\begin{aligned} Y_{i,t}^{(k)} = \beta_0 &+ \beta_1 \text{Foreign firm}_{i,t} + \beta_2 \log\left(1 + \text{Firm labor intensity}_{i,t}\right) \\ &+ \beta_3 \left[\text{Foreign firm}_{i,t} \times \log\left(1 + \text{Firm labor intensity}_{i,t}\right)\right] \\ &+ \boldsymbol{\gamma}' \mathbf{Z}_{i,t} + \alpha_t^{\text{Year}} + \alpha_{s(i,t)}^{\text{State}} + \alpha_{j(i,t)}^{\text{Industry}} + \varepsilon_{i,t}. \end{aligned}$$

Hypothesis 3 predicts that greater firm labor intensity attenuates the foreign-firm disadvantage. Accordingly, the interaction coefficient is expected to be positive ($\beta_3 > 0$).

The supply-chain interaction model tests whether supplier labor intensity conditions the relationship between foreign supplier reliance and subsidy allocation:

$$\begin{aligned} Y_{i,t}^{(k)} = \beta_0 &+ \beta_1 \text{Foreign supplier share}_{i,t-1} + \beta_2 \log\left(1 + \text{Supplier labor intensity}_{i,t}\right) \\ &+ \beta_3 \left[\text{Foreign supplier share}_{i,t-1} \times \log\left(1 + \text{Supplier labor intensity}_{i,t}\right)\right] \\ &+ \boldsymbol{\gamma}' \mathbf{Z}_{i,t} + \alpha_t^{\text{Year}} + \alpha_{s(i,t)}^{\text{State}} + \alpha_{j(i,t)}^{\text{Industry}} + \varepsilon_{i,t}. \end{aligned}$$

Hypothesis 4 predicts that supplier labor intensity strengthens the negative relationship between foreign supplier reliance and subsidy allocation. The corresponding interaction coefficient is therefore expected to be negative ($\beta_3 < 0$).

Foreign supplier share is lagged by one year to preserve the temporal ordering between supplier exposure and subsidy allocation. The baseline and interaction models are estimated using ordinary least squares with heteroskedasticity-robust standard errors. The dependent variables and labor-intensity measures are log-transformed to reduce skewness.

To provide substantive estimates on the original outcome scale, I also estimate Poisson pseudo-maximum-likelihood models using the untransformed subsidy outcomes and the same controls and fixed effects. Unlike direct back-transformations from the OLS models, PPML produces nonnegative conditional-mean predictions while accommodating zero-valued and highly skewed outcomes. I use these models to calculate average adjusted predictions under counterfactual U.S.-firm and foreign-firm status, averaging over the observed distribution of firm size, year, state, and industry.

# Results

Table 2 presents the baseline relationship between firm foreignness and industrial subsidy allocation among publicly traded firms in the United States.[8] Consistent with Hypothesis 1, foreign nationality is associated with a substantial disadvantage in subsidy allocation. In Column 1, foreign firms receive significantly fewer subsidies than U.S. firms ($\beta = -0.115$, $p < 0.01$), corresponding to approximately 11% fewer subsidies on average. In the joint specification reported in Column 3, the foreign-firm coefficient is substantially larger ($\beta = -0.785$, $p < 0.001$), implying that foreign firms receive about 54% fewer subsidies than otherwise comparable U.S. firms in the supplier-information sample.[9]

A similar pattern emerges for subsidy amounts. In Column 1, foreign firms receive roughly 63% smaller subsidy amounts than U.S. firms ($\beta = -0.995$, $p < 0.01$). The joint specification in Column 3 produces a much larger estimate ($\beta = -6.015$, $p < 0.001$). As noted above, however, Column 3 is estimated on the restricted supplier-information sample, so the increase in magnitude cannot be attributed solely to the inclusion of foreign supplier

[8] Appendix tables report robustness checks, including specifications without fixed effects, logit models for binary outcomes, and zero-inflated models to account for excess zeros. The substantive conclusions remain unchanged across these alternative specifications.

[9] Column 1 uses the full sample, whereas Columns 2 and 3 are restricted to firms for which supplier information is available. Firms in the supplier-information sample tend to be larger and receive more frequent and larger subsidies than firms in the full sample. The foreign-firm coefficients in Columns 1 and 3 are therefore not directly comparable: their difference may reflect the composition of the restricted supplier-information sample as well as the inclusion of foreign supplier share in Column 3. Accordingly, the larger coefficient in Column 3 should not be interpreted as the effect of controlling for supplier exposure alone.

Table 2: Firm Foreignness and Industrial Subsidy Allocation

| | 1<br>Full sample | 2<br>Supplier sample | 3<br>Joint model |
|---|---|---|---|
| | Log Count of Industrial Subsidies | | |
| Foreign firm | -0.115*** | | -0.785*** |
| | (0.026) | | (0.072) |
| Foreign supplier share$_{t-1}$ | | 0.097** | 0.097** |
| | | (0.034) | (0.034) |
| | Log Amount of Industrial Subsidies | | |
| Foreign firm | -0.995** | | -6.015*** |
| | (0.310) | | (1.332) |
| Foreign supplier share$_{t-1}$ | | 0.733* | 0.739* |
| | | (0.307) | (0.307) |
| Firm size | Yes | Yes | Yes |
| State FE | Yes | Yes | Yes |
| Year FE | Yes | Yes | Yes |
| Industry FE | Yes | Yes | Yes |

Note: Ordinary least squares estimates with robust standard errors in parentheses. All models include firm size, year, state, and 4-digit NAICS industry fixed effects. $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

share.[10] Together, these results provide strong support for the hypothesis that foreign firms face a systematic disadvantage in industrial subsidy allocation.

On the other hand, the relationship between supply-chain foreignness and subsidies differs markedly from the firm-nationality effect. Firms with a larger share of foreign suppliers receive significantly more subsidies and larger subsidy amounts. In both the supplier-only specification (Column 2) and the joint specification (Column 3), a one-unit increase in foreign supplier share is associated with a coefficient of 0.097 for subsidy counts ($p < 0.05$ and $p < 0.01$, respectively), implying approximately a 10% increase in the expected number of subsidies. Similarly, the estimated coefficients for subsidy amounts (0.733 and 0.739) indicate that firms with completely foreign rather than completely domestic supplier networks receive roughly 108–109% larger subsidy amounts, holding other factors constant.

[10]To address concerns about comparability across firms, I also estimate models using matching procedures that balance firms on observable characteristics. Specifically, I implement Coarsened Exact Matching and Generalized Full Matching to compare firms of similar size operating within the same state, industry, and year. The results remain consistent with the baseline findings: firms with foreign nationality receive less industrial support even among observationally comparable firms (Appendix C1). This suggests that the observed gap in subsidy allocation is not driven solely by observable firm characteristics.

These findings suggest that foreign suppliers are viewed differently from foreign firms. While foreign nationality may be politically costly because subsidy benefits can flow abroad, reliance on foreign suppliers does not appear to incur the same penalty. One possibility is that firms with observable global supplier networks are larger or more strategically positioned. The positive association may therefore reflect the characteristics of firms covered by the supplier data rather than a political preference for foreign sourcing itself.

Overall, the baseline results demonstrate that industrial subsidy allocation depends not simply on whether firms are globally integrated, but on how they are integrated. Foreign nationality is penalized, consistent with concerns about economic leakage and political reciprocity, whereas participation in global supply chains is not systematically discouraged. These findings indicate that foreign nationality carries a systematic penalty, but also suggest that this penalty is unlikely to be uniform across firms. The next section examines whether this penalty varies with firms' labor intensity, the empirical proxy for their employment-generating orientation.

## Substantive interpretation and robustness checks

To provide a substantive interpretation on the original outcome scale, Table 3 reports average adjusted predictions from Poisson pseudo-maximum-likelihood models. These predictions average over the observed distribution of firm size, year, state, and industry rather than representing a hypothetical firm with fixed characteristics. The results indicate that, holding the included covariates and fixed effects constant, U.S. firms are predicted to receive 0.471 subsidies per firm-year, compared with 0.194 for foreign firms, a difference of 0.277 awards, or approximately 58.8%. Predicted subsidy amounts exhibit an even larger disparity: U.S. firms are predicted to receive an average of \$1.766 million per firm-year, compared with \$0.396 million for foreign firms, a difference of approximately \$1.371 million, or 77.6%. These estimates reinforce the finding that foreign nationality is associated with lower industrial support and suggest that the disparity is especially pronounced for the amount of support received.

Because the PPML models estimate conditional means on the original outcome scale and are more sensitive to large subsidy awards, their estimated magnitudes differ from those obtained from the OLS models using $\log(1 + y)$ outcomes; the two approaches nevertheless produce the same substantive conclusion regarding the direction of the foreign-firm disadvantage.

The average adjusted predictions reveal substantial differences between U.S. and foreign

Table 3: Average adjusted subsidy outcomes by firm nationality

| Outcome | U.S. firms | Foreign firms | Difference | % difference |
|---|---|---|---|---|
| Subsidy count | 0.471 | 0.194 | -0.277 | -58.8% |
| Subsidy amount ($ millions) | 1.766 | 0.396 | -1.371 | -77.6% |

Note: The table reports average adjusted predictions from Poisson pseudo-maximum-likelihood models. For each firm-year, predictions are calculated under counterfactual U.S.-firm and foreign-firm status while retaining the observation's actual firm size, year, state, and four-digit NAICS industry. Predictions are then averaged across the estimation sample. Subsidy amounts are reported in millions of U.S. dollars. Percentage differences are calculated as (Foreign − U.S.)/U.S. × 100. Models use heteroskedasticity-robust standard errors.

firms. The PPML estimates predict 0.471 subsidies per firm-year for U.S. firms and 0.194 for foreign firms, implying that foreign firms receive approximately 58.8% fewer subsidies. The corresponding predicted subsidy amounts are $1.766 million for U.S. firms and $0.396 million for foreign firms, a difference of approximately 77.6%. Thus, the foreign-firm disadvantage is evident in both the number and total value of subsidies and is especially pronounced for subsidy amounts.

These results indicate that foreign nationality is associated not only with fewer subsidy awards but also with substantially lower total support. Because subsidy amount is measured for all firm-years, including those in which firms receive no subsidy, the amount difference reflects both access to subsidies and the value of the subsidies awarded. It should therefore not be interpreted solely as evidence that foreign firms receive smaller awards conditional on receiving support. Nevertheless, the larger disparity in total subsidy amounts suggests that the foreignness penalty is particularly consequential for the overall scale of government support.

As a robustness check, beyond formally recorded nationality, I also examine whether perceived national identity shapes subsidy allocation. In practice, policymakers may rely on observable cues, such as firm names, when evaluating firms' alignment with domestic interests. Using firm names to proxy for perceived foreignness, I find that firms with foreign-sounding names receive less industrial support, even among firms headquartered in the United States (Appendix C2). This pattern suggests that subsidy allocation is influenced not only by headquarters country, but also by how firms are perceived in terms of their national affiliation, reinforcing the role of political attribution in industrial policy decisions (Mutz, Mansfield, and Kim, 2020; Bankert, Powers, and Sheagley, 2022).

A related concern is that subsidy allocation is not solely a government decision. Firms themselves apply for support, and larger or more strategically positioned firms may be better positioned to identify relevant programs, meet application requirements, and coordinate with policymakers, reflecting their awareness of their own importance in the market. This raises a potential selection issue: observed disparities could partly reflect differences in application behavior rather than allocation decisions alone. However, because this analysis covers federal, state, and local subsidy programs rather than a single program with idiosyncratic eligibility rules, the influence of any one program's specific requirements or conditions is unlikely to critically undermine the broader pattern. If anything, the variation in program design across levels of government and the persistence of the foreignness penalty across this heterogeneous set of programs reinforces rather than undermines the argument: firms navigate this landscape strategically, but the resulting disparities in outcomes remain systematically patterned by recorded and perceived nationality.

For further robustness checks, I re-estimate the analysis using a linear probability model with binary subsidy receipt as the outcome (Appendix **??**), as well as a zero-inflated negative binomial model applied to the raw subsidy count (Appendix B5). Across these alternative specifications, the results remain largely consistent with the baseline findings, indicating that the observed foreignness penalty is not an artifact of the log-linear functional form or of how zero-subsidy observations are treated.

Taken together, the predicted outcomes confirm that foreign nationality is associated with systematic disadvantages in subsidy allocation, while also underscoring that foreign firms are not simply excluded. The results instead point to a more conditional pattern of subsidy allocation, in which the political costs of supporting foreign firms may be offset by signals of economically salient activity. This expectation suggests that the foreignness penalty should vary across firms rather than apply uniformly. The next section examines whether labor intensity, as a proxy for firms' employment orientation, conditions the relationship between foreign nationality and industrial subsidy allocation.

## Labor intensity and the foreignness penalty

While the baseline results establish a systematic foreignness penalty in industrial subsidy allocation, they also suggest that this penalty is not absolute. Building on the theoretical framework, this section argues that governments should be more willing to subsidize foreign firms when they generate greater domestic political returns. In particular, employment-intensive firms create visible economic benefits for local communities, increasing the electoral value of subsidies and offsetting concerns about economic leakage. If politicians balance these competing considerations, the foreignness penalty should weaken as firms become more labor intensive.

To evaluate this expectation, I extend the baseline models by interacting foreignness with measures of labor intensity at both the firm and supplier levels. Labor intensity is measured as the ratio of employees to revenue and captures the importance of labor relative to firms' economic scale. Because the argument centers on how domestic employment reshapes political incentives, I use labor intensity as an observable proxy for employment orientation. The measure does not reveal where employees are located, so the analysis tests whether the pattern is consistent with the proposed domestic-employment mechanism rather than directly measuring that mechanism.

The analysis also examines whether these dynamics extend beyond firms' own operations to their supply chains. Foreign supplier exposure indicates whether firms sourced inputs from foreign suppliers in the prior year (lagged), while supplier labor intensity captures the labor intensity of those supplier networks. As with the firm-level measure, supplier labor intensity does not identify where employees are located. All models include controls for firm size and fixed effects for state, year, and industry.

To evaluate whether employment orientation conditions subsidy allocation, Table 4 introduces firm- and supplier-level measures of labor intensity. Models 1 and 2 focus on nationality-based foreignness, while Models 3 and 4 examine supply-chain foreignness.

Model 1 largely replicates the baseline relationship. Foreign firms receive significantly fewer subsidies than comparable U.S. firms ($\beta = -0.116$, $p < 0.01$), corresponding to approximately an 11% reduction in expected subsidy counts. At the same time, labor intensity is strongly and positively associated with subsidy receipt ($\beta = 0.430$, $p < 0.01$). A one-unit increase in logged labor intensity is associated with roughly a 54% increase in expected subsidy counts, indicating that more labor-intensive firms are substantially more likely to receive industrial support.

Model 2 introduces the interaction between foreign nationality and labor intensity. The interaction coefficient is positive and statistically significant ($\beta = 0.638$, $p < 0.01$), indicating that the negative association between foreign nationality and subsidy receipt diminishes as firms become more labor intensive. Although foreign firms continue to face a baseline disadvantage ($\beta = -0.119$, $p < 0.01$), the positive interaction shows that this disadvantage narrows as logged labor intensity increases, providing support for Hypothesis 3. This pattern is consistent with the argument that an employment-generating orientation can make foreign firms more attractive recipients of industrial support, although the geographic mechanism remains unobserved.

Table 4: Moderator Effects of Labor Intensity

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| | Log Count of Industrial Subsidy | | | |
| Foreign Firm | -0.116** | -0.119** | | |
| | (0.036) | (0.036) | | |
| Labor Intensity (log) | 0.430*** | 0.247** | | |
| | (0.051) | (0.060) | | |
| Foreign Firm × Labor Intensity (log) | | 0.638*** | | |
| | | (0.109) | | |
| Foreign supplier share$_{t-1}$ | | | 0.098** | 0.095* |
| | | | (0.038) | (0.039) |
| Supplier Labor Intensity (log) | | | 0.945** | 0.935** |
| | | | (0.289) | (0.291) |
| Foreign supplier share$_{t-1}$ × Supplier labor intensity (log) | | | | 0.870 |
| | | | | (3.185) |
| Firm Size Control | Yes | Yes | Yes | Yes |
| State FE | Yes | Yes | Yes | Yes |
| Year FE | Yes | Yes | Yes | Yes |
| Industry FE | Yes | Yes | Yes | Yes |

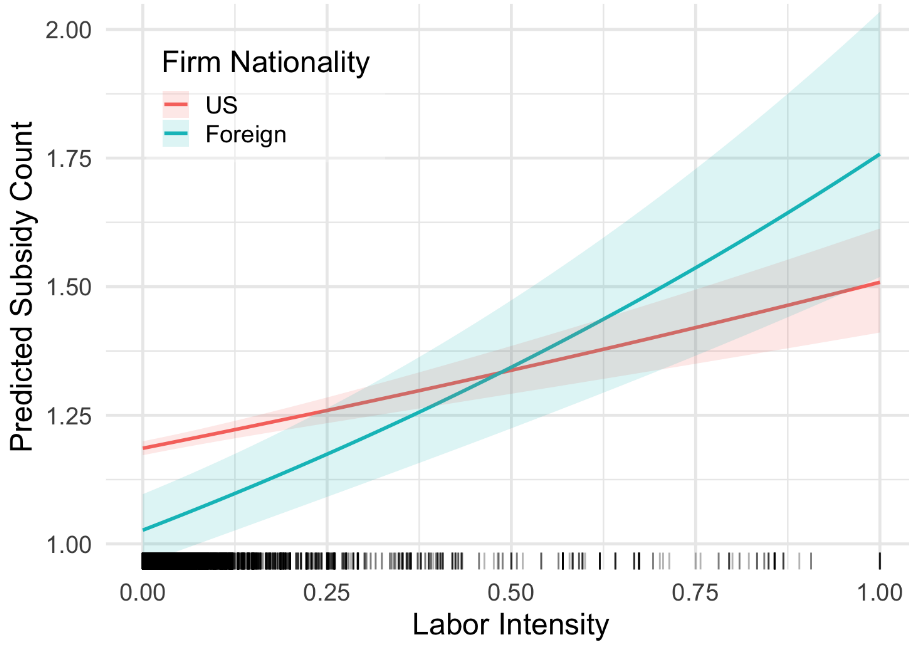
Notes: All models are OLS with robust standard errors.
$^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

For substantive interpretation, Figure 3 plots the predicted subsidy counts across the observed range of labor intensity. At relatively low levels of labor intensity (approximately the 25th percentile), foreign firms are predicted to receive significantly fewer subsidies than otherwise comparable U.S. firms. As labor intensity increases, however, the gap steadily narrows. By roughly the median level of labor intensity, the difference between foreign and U.S. firms becomes statistically indistinguishable from zero. The figure therefore illustrates how the foreignness penalty becomes progressively smaller as firms become more labor intensive.

Models 3 and 4 extend the analysis to supply chains. In Model 3, both foreign supplier exposure and supplier labor intensity are positively associated with subsidy receipt. However, Model 4 shows that the interaction between foreign supplier exposure and supplier labor intensity is not statistically significant. This suggests that, while firms connected to foreign suppliers and labor-intensive supply chains are not systematically penalized in isolation, there is no clear evidence that supplier labor intensity conditions the effect of foreign supplier exposure on subsidy allocation.

Taken together, these results highlight an asymmetry between firm- and supply-chain-level dynamics. At the firm level, labor intensity clearly moderates the foreignness penalty:

Figure 3: Predicted subsidy count by firm nationality and labor intensity




more labor-intensive foreign firms face a smaller disadvantage. At the supply-chain level, by contrast, the hypothesized moderating role of supplier labor intensity is not supported. This asymmetry aligns with the argument that employment-oriented activity at the subsidized firm carries more political weight than labor intensity embedded in upstream supply chains. Because neither measure identifies where employees are located, however, the findings should be interpreted as evidence about employment orientation rather than the geographic location of jobs.

# Conclusion

This study examines why governments subsidize foreign firms, and under what conditions firm nationality shapes subsidy allocation. Using firm-level data on U.S. industrial subsidies from 2000 to 2023, I show that governments selectively support foreign firms: while such firms face a systematic disadvantage relative to U.S. counterparts, this penalty is conditional. More labor-intensive foreign firms face weaker disadvantages, while firms' reliance on foreign suppliers is not associated with reduced support and is not conditioned by supplier labor intensity. The firm-level result is consistent with the argument that an employment-generating orientation can make foreign firms more politically defensible, although the available measure does not identify whether employment is located domestically or abroad.

More broadly, the results highlight a central tension in contemporary industrial policy. Governments increasingly rely on subsidies to promote competitiveness, technological leadership, and economic security in response to globalization and geopolitical rivalry. Yet globalization simultaneously complicates the implementation of these policies. Firms that are strategically important often operate across borders, meaning public resources may benefit foreign shareholders or foreign workers as well as domestic constituencies. Policymakers therefore confront competing incentives: supporting globally integrated firms may advance national economic objectives, but doing so can impose political costs when benefits appear to accrue abroad. The foreignness penalty identified in this study reflects how governments navigate this trade-off.

The findings contribute to several strands of research while speaking directly to contemporary debates on the resurgence of industrial policy. First, they advance work on the political economy of industrial policy by shifting attention from sectoral and regional targeting to within-industry allocation across firms. While existing studies emphasize how governments direct support toward strategic industries or locations (Rickard, 2018; Slattery, 2023), this study shows that policymakers also differentiate among firms based on nationality and employment orientation. Second, the results contribute to emerging scholarship on economic statecraft and geoeconomic competition by clarifying how governments manage globalization under conditions of interdependence (Lane, 2020; Juhász, Lane, and Rodrik, 2024). Rather than simply favoring nationally based firms or reshoring production wholesale, industrial policy appears to reward firm characteristics associated with politically salient economic activity, even when firms remain globally integrated. At the same time, the absence of systematic penalties tied to supplier characteristics suggests limits to governments' ability, or willingness, to discipline complex global supply chains. Third, the study extends firm-centered approaches in international political economy and New New Trade Theory by identifying firm nationality and labor intensity as key correlates of state support (Melitz, 2003; Osgood, 2019; Kim and Osgood, 2019). Firm heterogeneity shapes not only trade politics but also how governments allocate industrial policy resources in practice.

The analysis also points to an important conceptual implication: governments differentiate between access to industrial policy and the intensity of support. Foreign firms are not systematically excluded from subsidy programs, but they receive significantly smaller amounts when they do receive assistance. This distinction suggests that policymakers manage globalization partly by rationing public resources, balancing economic objectives with political considerations about national alignment. Moreover, the evidence that foreign-sounding firm names are associated with lower subsidy allocation, even among domestically headquartered firms, indicates that perceptions of national identity, not only observed headquarters country, shape policy outcomes.

These findings open several avenues for future research. Comparative studies could examine whether similar foreignness penalties emerge across institutional contexts and political systems. Additional work could explore how lobbying, geopolitical competition, or national security concerns interact with firm characteristics to shape industrial policy decisions. Research on the long-term consequences of subsidies for firm performance, innovation, and supply chain restructuring would also deepen understanding of industrial policy effectiveness in a globalized economy.

Ultimately, this study reframes industrial policy as a tool for governing interdependence rather than protecting purely national industries. Governments face the challenge of supporting firms that are globally embedded while maintaining domestic political legitimacy. The evidence presented here shows that subsidy allocation is associated not only with firm nationality, but also with firms' employment orientation. Future research using geographically disaggregated employment data can more directly test whether the domestic location of jobs accounts for this pattern. As globalization continues to reshape production networks and national economies, understanding how states balance these competing pressures will be central to explaining the future of industrial policy and economic statecraft.

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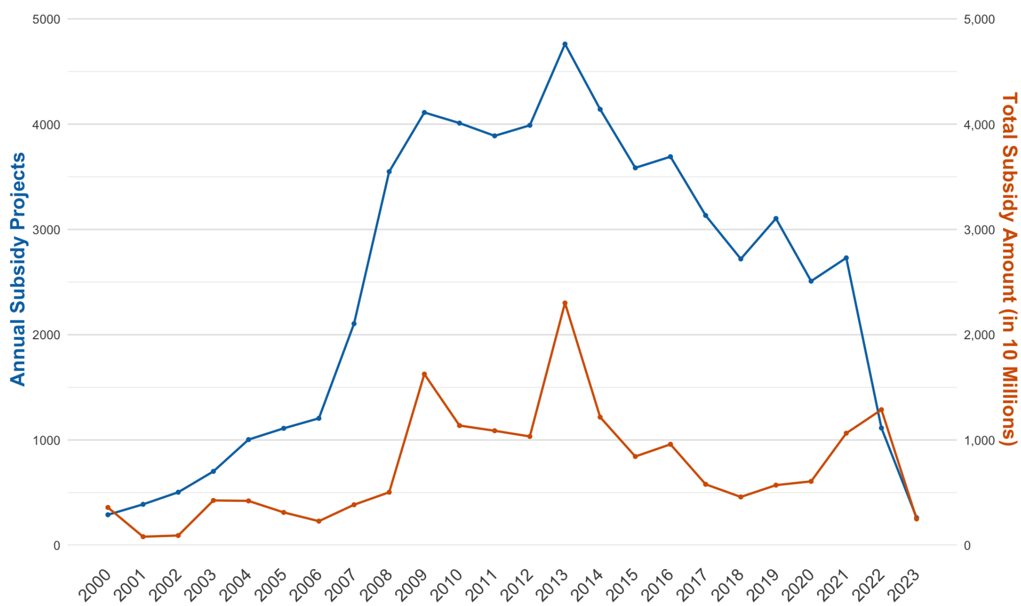
# Appendix A: Descriptive statistics

Table A1: Summary statistics

| | Count | Mean | SD | Min | Max |
|---|---|---|---|---|---|
| Annual subsidy count | 151338 | 0.38 | 2.62 | 0.00 | 196.00 |
| Annual subsidy amount | 151338 | $1.16 \times 10^6$ | $4.00 \times 10^7$ | 0.00 | $8.83 \times 10^9$ |
| U.S. firm | 151338 | 0.77 | 0.42 | 0.00 | 1.00 |
| Foreign firm | 151338 | 0.23 | 0.42 | 0.00 | 1.00 |
| Foreign supplier share | 20996 | 0.04 | 0.18 | 0.00 | 1.00 |
| Firm labor intensity | 151338 | 0.01 | 0.02 | 0.00 | 1.00 |
| Supplier labor intensity | 20996 | 0.01 | 0.05 | 0.00 | 1.00 |
| Annual revenue | 151338 | $3.71 \times 10^3$ | $1.68 \times 10^4$ | 0.00 | $6.48 \times 10^5$ |
| Number of employees | 151338 | 10.03 | 44.06 | 0.00 | 2300.00 |

Note: The regression analyses transform subsidy count and subsidy amount as $\log(1+y)$ and log-transform both labor-intensity measures. Annual revenue and employee count are standardized and combined into a composite firm-size control.

Figure A1: Annual count and amount of industrial subsidy projects




Note: Subsidies are assigned to the year in which they were implemented. Because implementation may occur after an award is announced, the figure may reflect a lag between subsidy decisions and recorded subsidy activity.

Table A2: Balance table

| | Non-subsidized Firms | Subsidized Firms |
|---|---|---|
| U.S. firm | 0.76 | 0.846 |
| Foreign firm | 0.238 | 0.154 |
| Foreign supplier share | 0.039 | 0.032 |
| Firm labor intensity | 0.011 | 0.004 |
| Supplier labor intensity | 0.033 | 0.004 |
| Annual revenue | 2125.09 | 19222.45 |
| Number of employees | 6.208 | 47.442 |

Note: Subsidized firms have higher revenue and more employees than non-subsidized firms, indicating their greater scale of operations.

Table A3: Industrial subsidy awards by industry (NAICS two-digit)

| **Code** | **Industry** | **Awards** | **Amount ($B)** |
|---|---|---|---|
| 33 | Manufacturing (Metal, Machinery, Electronics) | 15,765 | 63.74 |
| 32 | Manufacturing (Wood, Paper, Petroleum, Chemicals) | 7,871 | 21.02 |
| 52 | Finance and Insurance | 5,053 | 8.36 |
| 51 | Information | 4,613 | 13.10 |
| 31 | Manufacturing (Food, Textiles, Apparel) | 3,984 | 4.43 |
| 45 | Retail Trade | 2,605 | 5.19 |
| 22 | Utilities | 1,975 | 25.24 |
| 48 | Transportation and Warehousing | 1,971 | 4.95 |
| 44 | Retail Trade | 1,894 | 0.97 |
| 99 | Nonclassifiable | 1,838 | 0.38 |
| 42 | Wholesale Trade | 1,408 | 0.55 |
| 21 | Mining, Quarrying, and Oil & Gas Extraction | 1,223 | 4.34 |
| 49 | Transportation and Warehousing | 1,095 | 0.53 |
| 54 | Professional, Scientific, and Technical Services | 903 | 1.66 |
| 53 | Real Estate and Rental and Leasing | 831 | 2.18 |
| 72 | Accommodation and Food Services | 800 | 0.66 |
| 62 | Health Care and Social Assistance | 604 | 0.61 |
| 56 | Administrative and Support Services | 449 | 0.13 |
| 71 | Arts, Entertainment, and Recreation | 242 | 0.36 |
| 23 | Construction | 160 | 0.08 |
| 61 | Educational Services | 97 | 0.11 |
| 81 | Other Services | 47 | 0.01 |
| 11 | Agriculture, Forestry, Fishing, and Hunting | 31 | 0.04 |
| 92 | Public Administration | 0 | 0.00 |

Note: Industries are classified using two-digit NAICS codes. The table reports the total number of subsidy awards and cumulative subsidy amount received by firms in each industry over the sample period.

## A.1. Distribution of dependent variables

Figure A2: Distribution of subsidy count (recipients only)

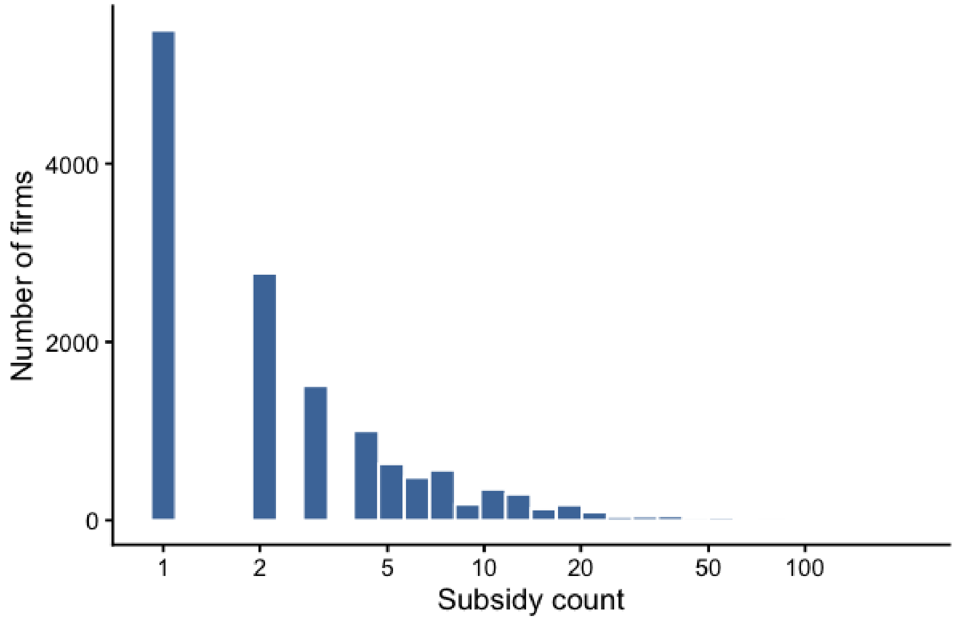
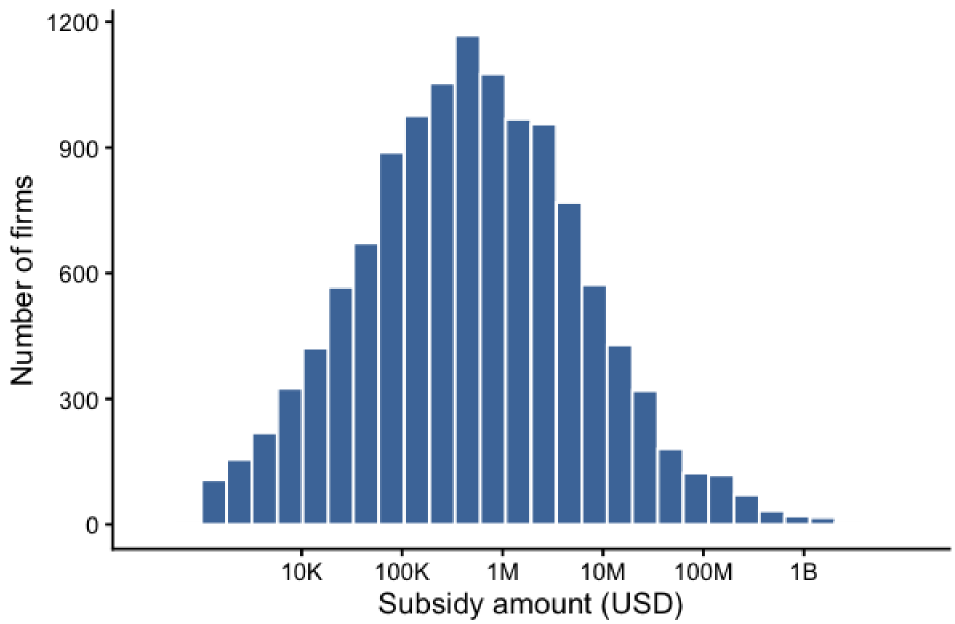



Figure A3: Distribution of subsidy amount (log scale, recipients only)




Note: Histograms are restricted to firms receiving at least one subsidy. The x-axes are displayed on a logarithmic scale to improve visualization of highly skewed distributions.

## A.2. Distribution of industrial subsidies by firm nationality

Figure A4: Number of firms receiving industrial subsidies, by headquarters country

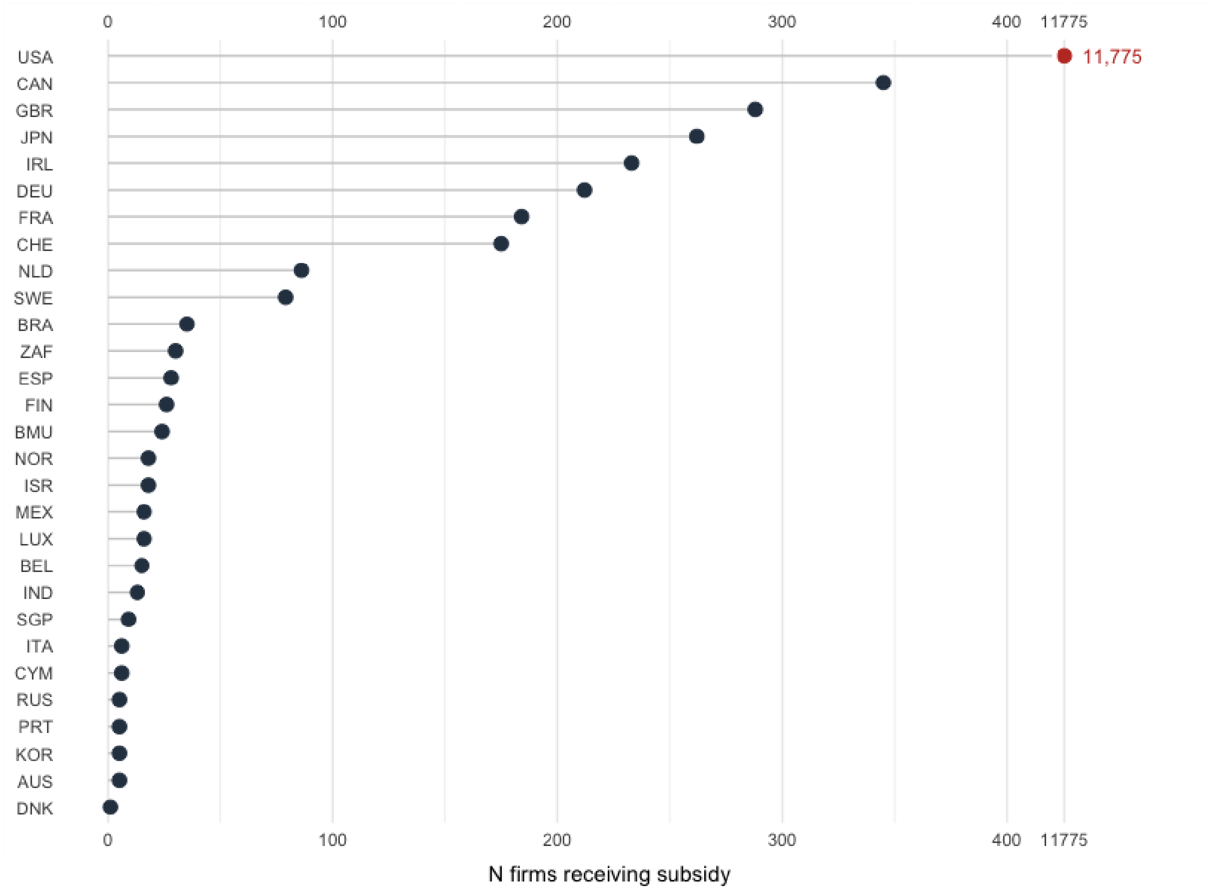
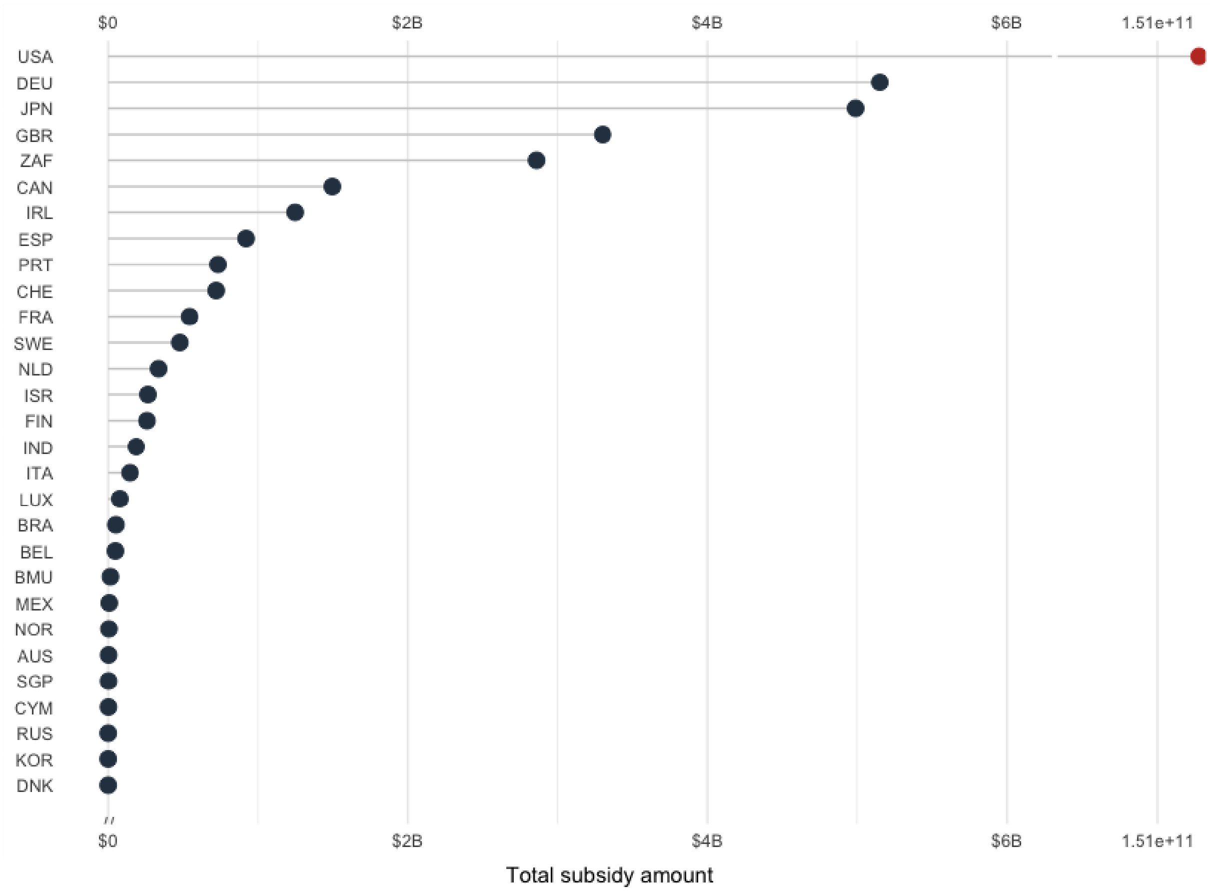


Figure A5: Total industrial subsidy amount, by headquarters country




Note: Countries refer to the location of firms' global headquarters. The figures report (a) the number of firms receiving at least one industrial subsidy and (b) the cumulative dollar value of subsidies awarded over the sample period.

## A.3. Distribution of labor-intensity measures

Figure A6: Distribution of average labor intensity

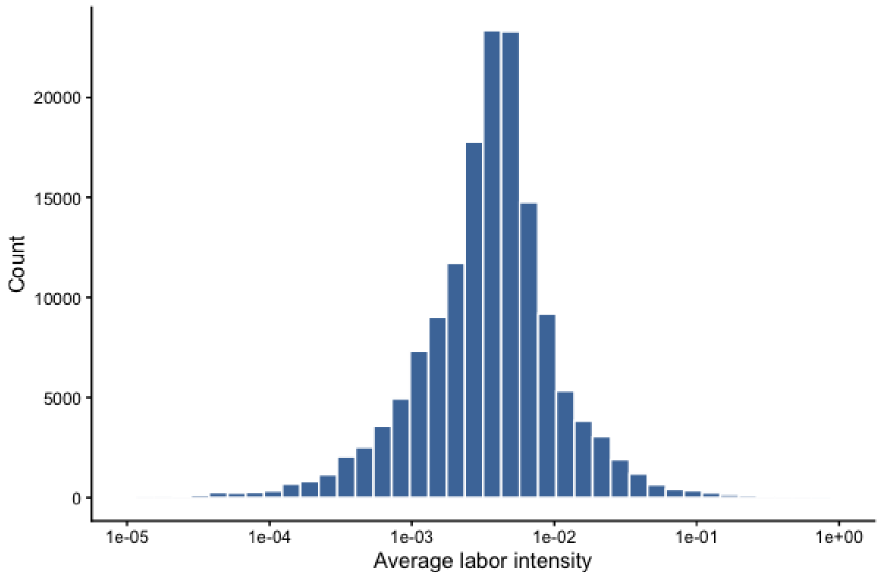
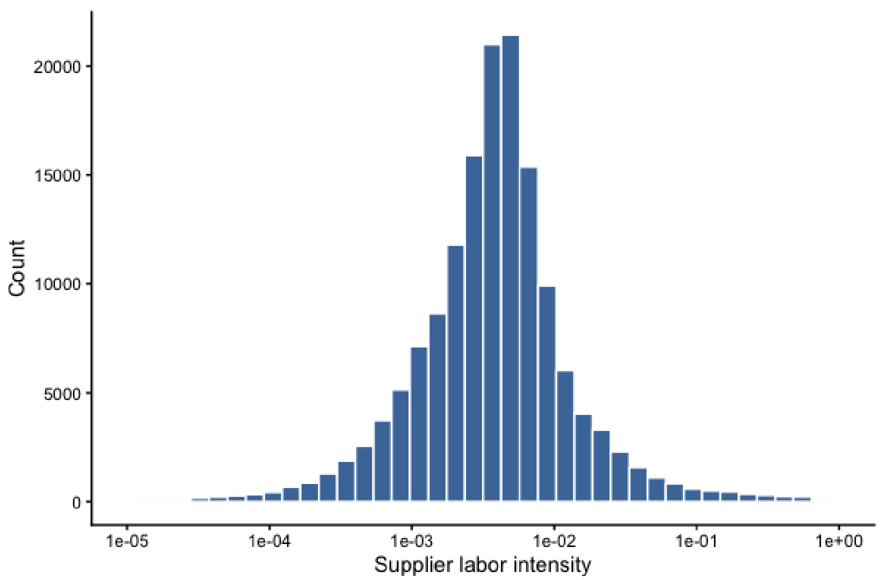



Figure A7: Distribution of supplier labor intensity




Note: The figures display the distributions of firm-level average labor intensity and supplier labor intensity. Both x-axes are presented on a logarithmic scale to improve visualization of the highly right-skewed distributions. Labor intensity is measured as the ratio of employees to annual sales for focal firms and their suppliers, respectively. The measures do not identify employees' geographic location.

# Appendix B: Robustness checks

## Models with firm ID-clustered standard errors

Table B1: Robustness check: Firm-clustered standard errors

| | 1 | 2 | 3 |
|---|---|---|---|
| | Log Count of Subsidy | | |
| Foreign firm | -0.115 | | -0.785*** |
| | (0.076) | | (0.124) |
| Foreign supplier share$_{t-1}$ | | 0.097 | 0.097 |
| | | (0.085) | (0.085) |
| | Log Amount of Subsidy | | |
| Foreign firm | -0.955** | | -6.015*** |
| | (0.310) | | (1.332) |
| Foreign supplier share$_{t-1}$ | | 0.733* | 0.739* |
| | | (0.307) | (0.307) |
| Firm size | Yes | Yes | Yes |
| State FE | Yes | Yes | Yes |
| Year FE | Yes | Yes | Yes |
| Industry FE | Yes | Yes | Yes |

Notes: Estimates with standard errors clustered by firm in parentheses.
All models include firm size, year, state, and 4-digit NAICS industry fixed effects.
$^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

As a robustness check, Table B1 re-estimates the baseline specifications with standard errors clustered by firm (`gvkey`) rather than heteroskedasticity-robust standard errors, to account for serial correlation in subsidy receipt within firms over time. The foreign nationality penalty remains negative and significant in the fully specified models (column 3) for both subsidy count and subsidy amount, and the amount specification remains significant even without the supplier-exposure control (column 1). The count specification in column 1 loses significance under firm clustering, suggesting that some of the precision in the main-text estimate depends on the assumed error structure; the coefficient magnitude, however, is materially unchanged. The foreign supplier share coefficient is directionally consistent with the main results throughout, reaching conventional significance in the amount models but not in the count models under firm-clustered errors. Overall, the core finding that foreign nationality is associated with reduced subsidy receipt, particularly once supplier embeddedness remains significantly and negatively associated.

## Fixed effects models

Table B2: Robustness check: Foreign firms across fixed-effects specifications

| | Log Count of Subsidy | | | Log Amount of Subsidy | | |
|---|---|---|---|---|---|---|
| Foreign firm | -0.105** | -0.122** | -0.115** | -0.895** | -0.989** | -0.955** |
| | (0.002) | (0.036) | (0.036) | (0.020) | (0.305) | (0.310) |
| Firm size | Yes | Yes | Yes | Yes | Yes | Yes |
| State FE | No | Yes | Yes | No | Yes | Yes |
| Year FE | No | Yes | Yes | No | Yes | Yes |
| NAICS 4-digit FE | No | No | Yes | No | No | Yes |

+ $p < 0.10$, * $p < 0.05$, ** $p < 0.01$

Table B3: Robustness check: Foreign supplier share across fixed-effects specifications

| | Log Count of Subsidy | | | Log Amount of Subsidy | | |
|---|---|---|---|---|---|---|
| Foreign supplier share$_{t-1}$ | -0.275** | -0.006 | 0.097* | -1.892** | -0.104 | 0.733* |
| | (0.040) | (0.039) | (0.038) | (0.315) | (0.308) | (0.307) |
| Firm size | Yes | Yes | Yes | Yes | Yes | Yes |
| State FE | No | Yes | Yes | No | Yes | Yes |
| Year FE | No | Yes | Yes | No | Yes | Yes |
| NAICS 4-digit FE | No | No | Yes | No | No | Yes |

+ $p < 0.10$, * $p < 0.05$, ** $p < 0.01$

Tables B2 and B3 test the robustness of the foreignness penalty using progressively saturated fixed-effects specifications. Foreign nationality remains significantly negatively associated with subsidy count and amount across all specifications, with stable coefficient magnitudes. Foreign supplier exposure, however, is sensitive to specification: its significant negative association with subsidies in the unconditional model disappears with state and year fixed effects and reverses to significantly positive once industry fixed effects are added. This suggests the foreignness penalty is robust for nationality-based foreignness but not for supply-chain-based foreignness, likely reflecting unobserved geographic and industry composition rather than a stable firm-level effect.

## Models with binary outcome

Table B4: Robustness check: Binary subsidy receipt

| | Subsidy receipt (binary) | | |
|---|---|---|---|
| Foreign firm | -0.070** | | -0.368** |
| | (0.024) | | (0.097) |
| Foreign supplier share$_{t-1}$ | | 0.057* | 0.057* |
| | | (0.022) | (0.022) |
| Firm size | Yes | Yes | Yes |
| State FE | Yes | Yes | Yes |
| Year FE | Yes | Yes | Yes |
| NAICS 4-digit FE | Yes | Yes | Yes |

+ $p < 0.10$, * $p < 0.05$, ** $p < 0.01$

Table B4 reports a linear probability model robustness check using a binary indicator for whether a firm received any subsidy, with state, year, and 4-digit NAICS fixed effects included throughout. Foreign nationality is significantly negatively associated with subsidy receipt whether entered alone or alongside foreign supplier exposure, and the effect strengthens substantially when both measures are included jointly. Foreign supplier exposure, by contrast, is positively associated with subsidy receipt in both specifications, consistent with the sign reversal observed in the count and amount models once full fixed effects are included. These results confirm that the foreignness penalty documented in the main analysis is not an artifact of the count or log-amount functional form, and that nationality- and supply-chain-based foreignness continue to diverge in direction even under a linear probability specification.

## Zero-inflated models

Table B5: Robustness check: Zero-inflated negative binomial models

| | Subsidy Count | | |
|---|---|---|---|
| | (1) | (2) | (3) |
| Foreign firm | -1.022+ | | -2.602** |
| | (0.573) | | (0.863) |
| Foreign supplier share$_{t-1}$ | | 0.156 | 0.161 |
| | | (0.169) | (0.169) |
| Firm size | Yes | Yes | Yes |
| State FE | Yes | Yes | Yes |
| Year FE | Yes | Yes | Yes |
| NAICS 4-digit FE | Yes | Yes | Yes |

Notes: Entries report coefficients from the count equation of zero-inflated negative binomial models.
+ $p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table B5 further examines the robustness of the baseline findings using zero-inflated negative binomial models to account for excess zeros in subsidy counts. Foreign nationality remains negatively associated with subsidy receipt across specifications and continues to be statistically significant in the full model, although the estimated effect is larger than in the baseline models. By contrast, the positive association between foreign supplier exposure and subsidy counts is no longer statistically significant. These results suggest that the foreignness penalty is robust for nationality-based foreignness but less so for supply-chain-based foreignness when accounting for excess zeros in the distribution of subsidy counts.

# Appendix C: Additional analyses

This appendix reports additional analyses examining the robustness and scope of the relationship between firm foreignness and industrial subsidy allocation. The analyses address observable differences between foreign and U.S. firms, perceived foreignness, industry trade exposure, sectoral heterogeneity, and variation across levels of government.

## Matching analysis

Table C1: Coarsened exact matching and generalized full matching

| | Log Count of Subsidy | |
|---|---|---|
| | CEM | GFM |
| Foreign firm | -0.074*** | -0.126*** |
| | (0.005) | (0.026) |
| Firm size | Yes | Yes |
| State FE | Yes | Yes |
| Year FE | Yes | Yes |
| Industry FE | Yes | Yes |

Standard errors are clustered at the firm level.
$+p < 0.10$, $*p < 0.05$, $**p < 0.01$, $***p < 0.001$

First, to reduce confounding and improve comparability across firms, I employ matching methods to balance the sample prior to estimation. Using Coarsened Exact Matching (CEM) and Generalized Full Matching (GFM) (Ho et al., 2007), I compare firms of similar size operating within the same state, industry, and year. These procedures ensure that differences in subsidy allocation are evaluated among observationally comparable firms. As reported in Table C1, the results remain consistent with the baseline findings: even among matched firms, foreign firms receive significantly less industrial support than U.S. firms. This provides additional evidence that the foreignness penalty is not driven solely by observable firm characteristics.

## Perceived foreignness and firm names

Second, I examine whether perceived foreignness influences subsidy allocation independently of firms' observed headquarters country. Firm nationality is often inferred through observable cues, such as company names, rather than recorded headquarters country alone. To capture this dimension, I classify firm names as English-sounding or foreign-sounding using OpenAI-based language detection methods combined with additional linguistic classification techniques. In the sample, approximately 13% of firms have foreign-sounding names, while 87% have English-sounding names. Notably, more than one-quarter of firms with English-sounding names are not headquartered in the United States, indicating that perceived nationality does not perfectly align with headquarters country.

Table C2: Foreign-sounding firm names and subsidy allocation

| | Full sample | | U.S. firms only | |
|---|---|---|---|---|
| | Log Count | Log Amount | Log Count | Log Amount |
| Foreign-sounding firm | -0.016** | -0.193** | -0.004+ | -0.075** |
| | (0.002) | (0.022) | (0.002) | (0.029) |
| Firm size | Yes | Yes | Yes | Yes |
| State FE | Yes | Yes | Yes | Yes |
| Year FE | Yes | Yes | Yes | Yes |
| Industry FE | Yes | Yes | Yes | Yes |

Notes: All models include firm size and state, year, and industry fixed effects.
Heteroskedasticity-robust standard errors are reported in parentheses.
$+p < 0.10$, $*p < 0.05$, $**p < 0.01$.

As shown in Table C2, firms with foreign-sounding names receive significantly less industrial support. In the full sample, these firms receive fewer subsidies and smaller subsidy amounts. The pattern persists even when restricting the analysis to U.S. firms, where foreign-sounding names remain associated with lower subsidy allocation. These results suggest that perceived national identity, independent of observed headquarters country, may influence how firms are evaluated as recipients of government support. More broadly, the findings indicate that the foreignness penalty operates partly through perceptual cues that shape judgments about firms' national alignment. Furthermore, these findings complement the labor-intensity results by showing that both economic contributions and perceived national identity shape subsidy allocation.

## Export and import exposure

Table C3: Industrial subsidies and industry export/import exposure

| | Log Count of Subsidy | | Log Amount of Subsidy | |
|---|---|---|---|---|
| | (1) | (2) | (3) | (4) |
| Export | 0.050** | 0.015** | 0.311** | 0.107** |
| | (0.008) | (0.005) | (0.051) | (0.033) |
| Import | 0.043** | -0.017** | 0.180** | -0.182** |
| | (0.009) | (0.005) | (0.057) | (0.037) |
| Industry size control | No | Yes | No | Yes |
| NAICS two-digit FE | Yes | Yes | Yes | Yes |
| Year FE | Yes | Yes | Yes | Yes |
| Observations | 5,979 | 5,979 | 5,979 | 5,979 |

Note: The unit of analysis is the NAICS four-digit industry-year.
Export and import exposure are measured at the industry-year level.
Models include year and NAICS two-digit fixed effects.
Heteroskedasticity-robust standard errors are reported in parentheses.
$+p < 0.10$, $*p < 0.05$, $**p < 0.01$.

## Tradable-goods and service-sector firms

Table C4: Industrial subsidies for tradable-goods and service-sector firms

| | Tradable Goods | | | | Services | | | |
|---|---|---|---|---|---|---|---|---|
| | Log Count | | Log Amount | | Log Count | | Log Amount | |
| Foreign firm | 0.119** | | 1.298** | | -0.117** | | -1.318** | |
| | (0.020) | | (0.210) | | (0.024) | | (0.326) | |
| Foreign supplier share$_{t-1}$ | | -0.001 | | -0.128 | | 0.033 | | 1.099* |
| | | (0.030) | | (0.386) | | (0.037) | | (0.510) |
| Firm size | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| State FE | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| Year FE | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| Industry FE | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes |

+ $p < 0.10$, * $p < 0.05$, ** $p < 0.01$
Note: Tradable goods include NAICS 11, 21, and 31–33, covering primary and manufacturing industries. Services include NAICS 51–56, 61, and 62. All models include firm size and state, year, and industry fixed effects. Standard errors are heteroskedasticity robust.